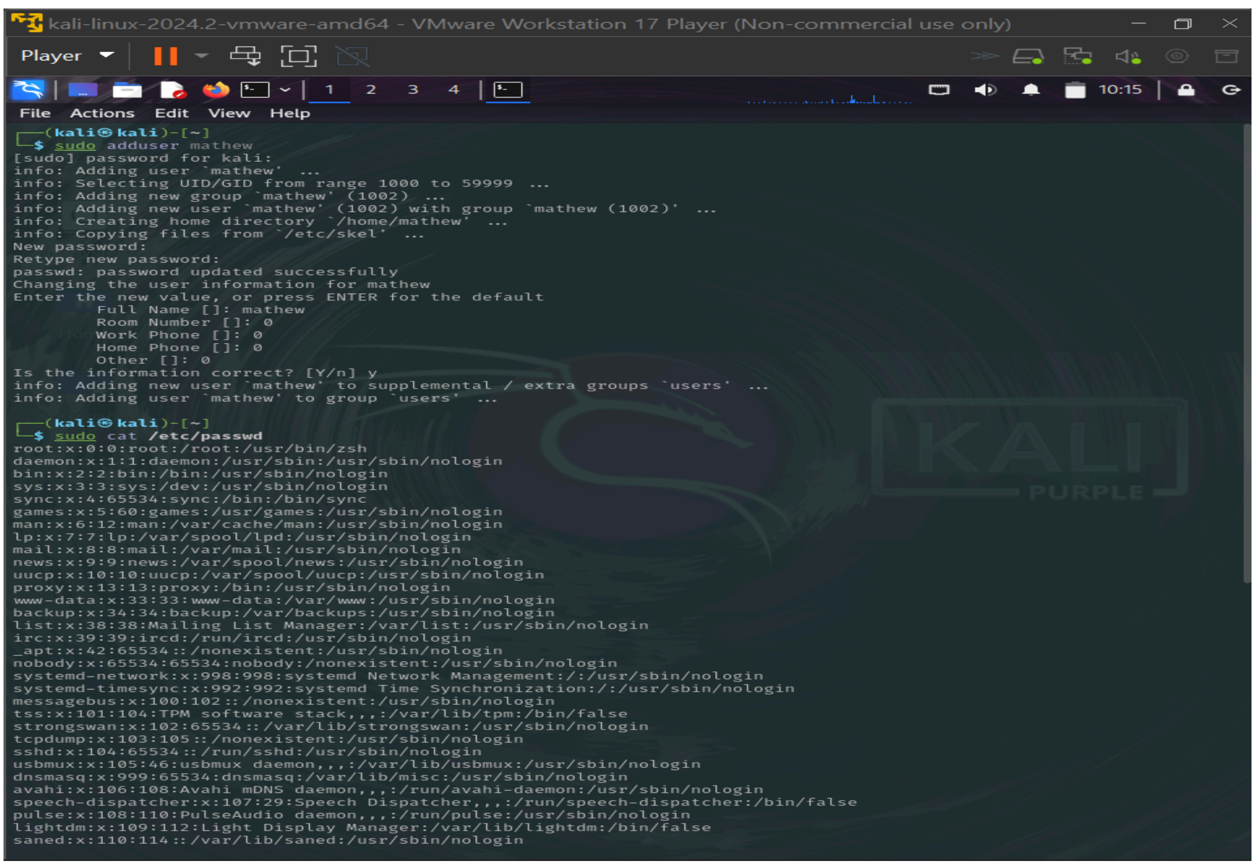
PASSWORD CRACKING WITH JOHN THE RIPPER : A DEEP DIVE INTO SINGLE, WORDLIST, INCREMENTAL AND DEFAULT MODES

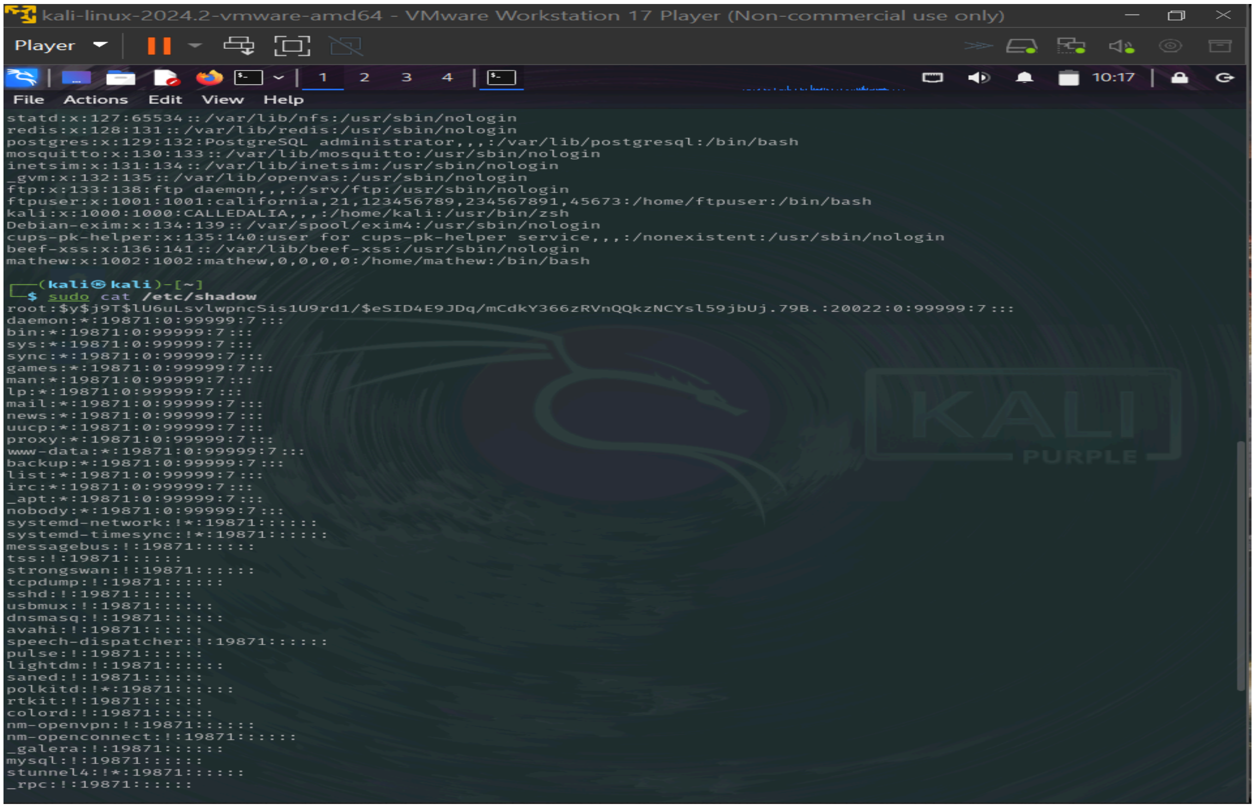
Tools : KALI LINUX , JOHN THE RIPPER

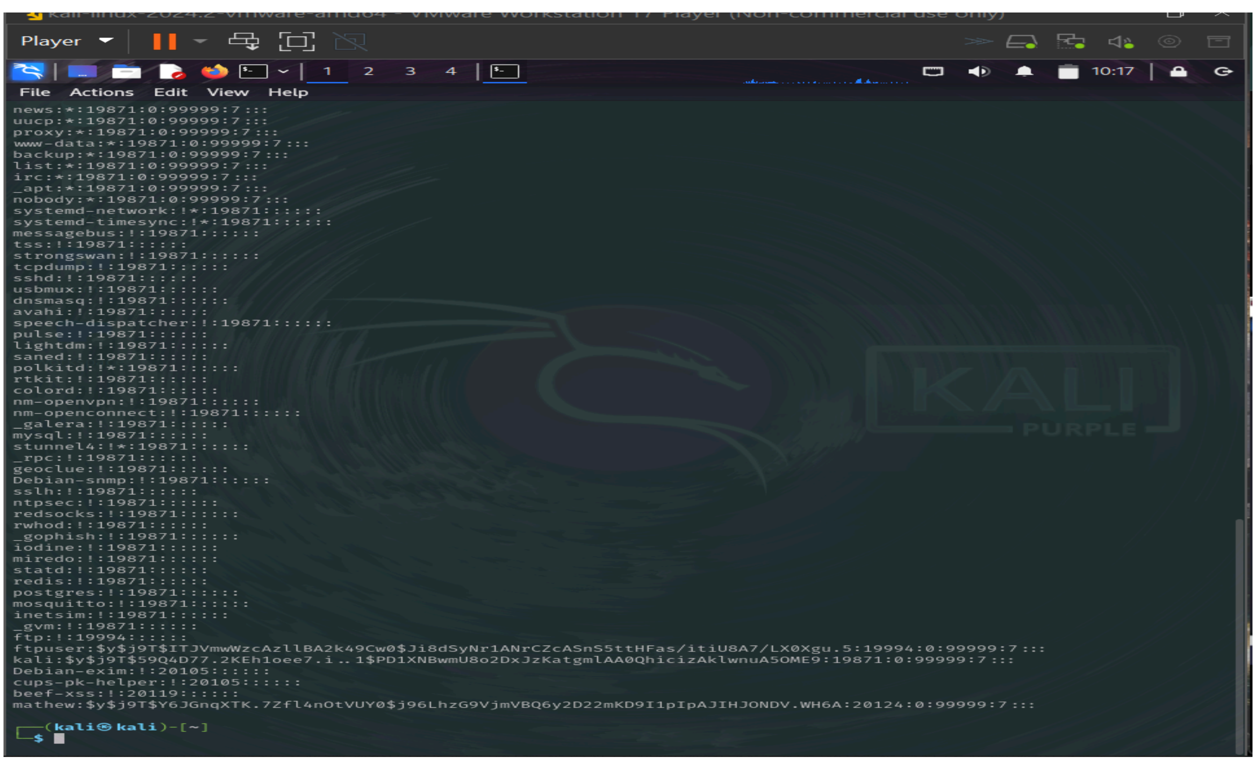
Overview

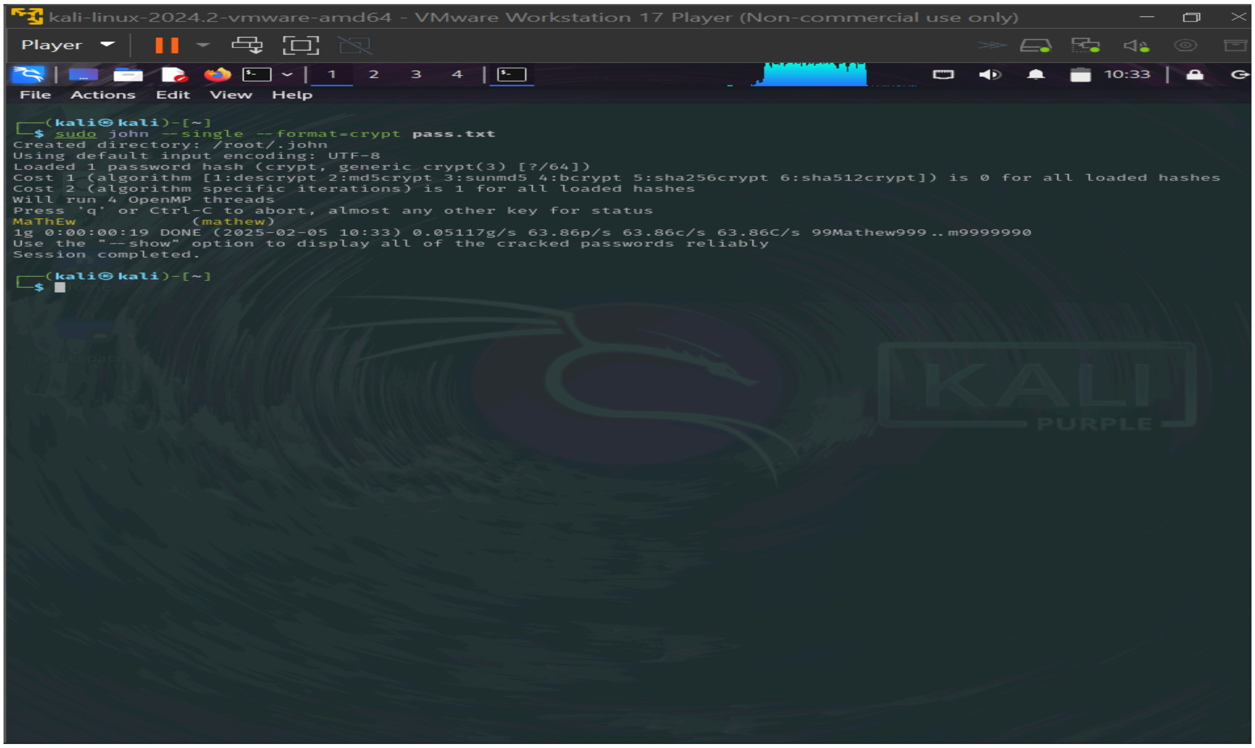
This session provides an in-depth exploration of John the Ripper, a powerful offline password-cracking tool. We will cover the fundamentals of password storage, the structure of password files in Linux, and demonstrate how to use John the Ripper in Single Crack, Wordlist, and Incremental modes. Additionally, we will discuss the default mode, which combines these strategies for efficient password cracking.

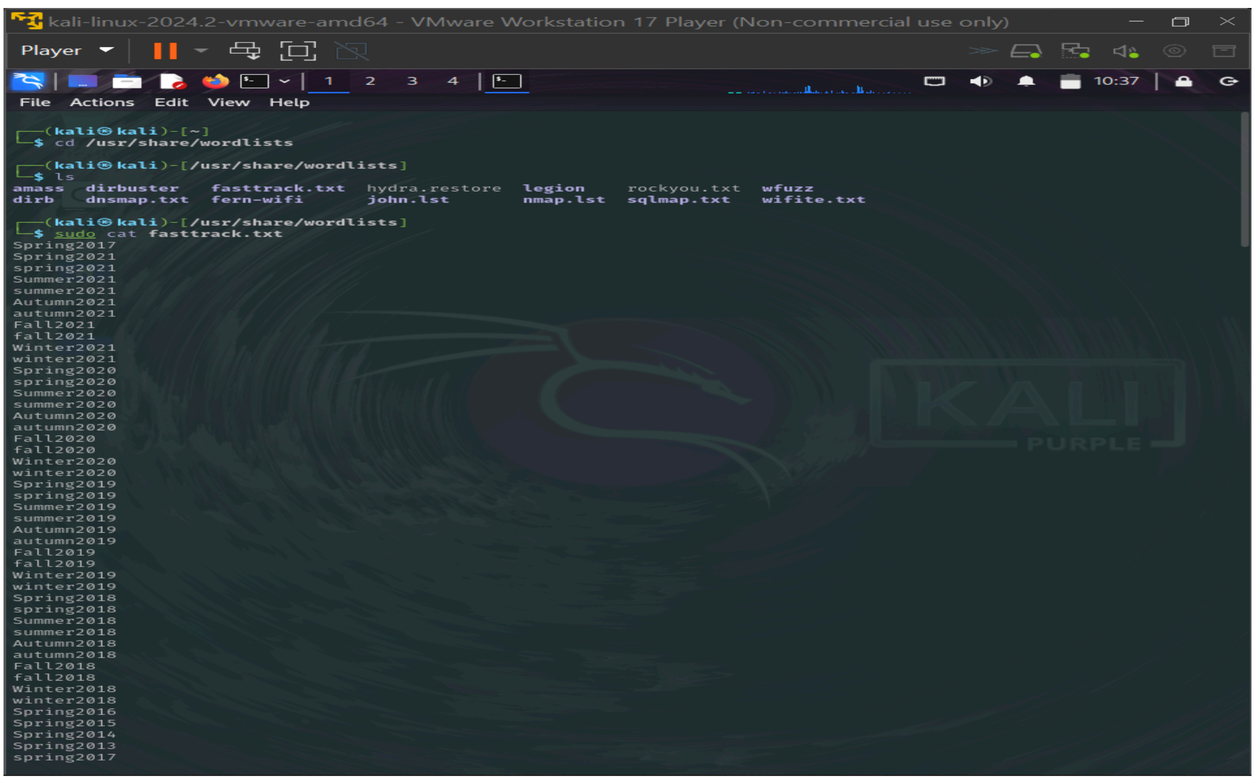
Input from kali :

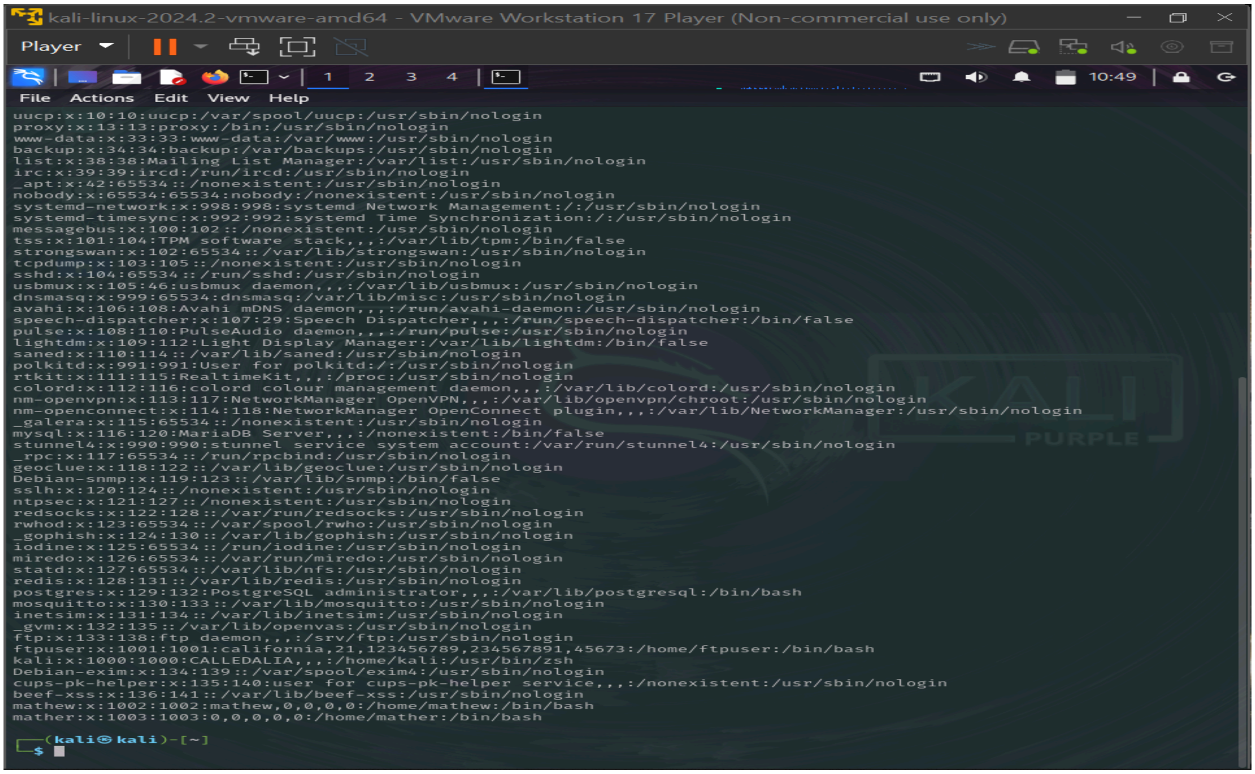


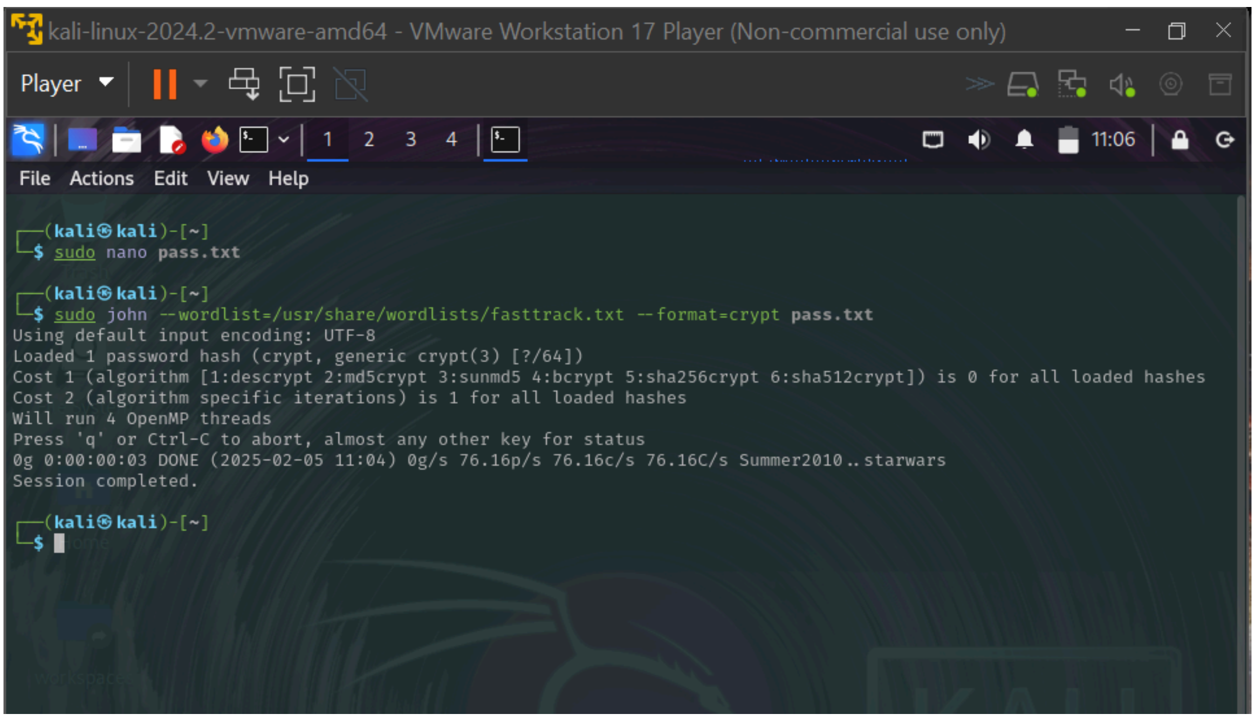

kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
10:17
File Actions Edit View Help
statd:x:127:65534::/var/lib/nfs:/usr/sbin/nologin
redis:x:128:131::/var/lib/redis:/usr/sbin/nologin
postgres:x:129:132:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
mosquitto:x:130:133::/var/lib/mosquitto:/usr/sbin/nologin
inetsim:x:131:134::/var/lib/inetsim:/usr/sbin/nologin
_gvm:x:132:135::/var/lib/openvas:/usr/sbin/nologin
ftp:x:133:138:ftp daemon,,,:/srv/ftp:/usr/sbin/nologin
ftpuser:x:1001:1001:california,21,123456789,234567891,45673:/home/ftpuser:/bin/bash
kali:x:1000:1000:CALLEDALIA,,,:/home/kali:/usr/bin/zsh
Debian-exim:x:134:139::/var/spool/exim4:/usr/sbin/nologin
cups-pk-helper:x:135:140:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
beef-xss:x:136:141::/var/lib/beef-xss:/usr/sbin/nologin
mathew:x:1002:1002:mathew,0,0,0,0:/home/mathew:/bin/bash
┌──(kali㉿kali)-[~]
└─$ sudo cat /etc/shadow
root:yj9T$lU6uLsvlwpncSis1U9rd1/$eSID4E9JDq/mCdkY366zRVnQQkzNCYsl59jbUj.79B.:20022:0:99999:7:::
daemon:*:19871:0:99999:7:::
bin:*:19871:0:99999:7:::
sys:*:19871:0:99999:7:::
sync:*:19871:0:99999:7:::
games:*:19871:0:99999:7:::
man:*:19871:0:99999:7:::
lp:*:19871:0:99999:7:::
mail:*:19871:0:99999:7:::
news:*:19871:0:99999:7:::
uucp:*:19871:0:99999:7:::
proxy:*:19871:0:99999:7:::
www-data:*:19871:0:99999:7:::
backup:*:19871:0:99999:7:::
list:*:19871:0:99999:7:::
irc:*:19871:0:99999:7:::
_apt:*:19871:0:99999:7:::
nobody:*:19871:0:99999:7:::
systemd-network:!*:19871::::::
systemd-timesync:!*:19871::::::
messagebus:!:19871::::::
tss:!:19871::::::
strongswan:!:19871::::::
tcpdump:!:19871::::::
sshd:!:19871::::::
usbmux:!:19871::::::
dnsmasq:!:19871::::::
avahi:!:19871::::::
speech-dispatcher:!:19871::::::
pulse:!:19871::::::
lightdm:!:19871::::::
saned:!:19871::::::
polkitd:!*:19871::::::
rtkit:!:19871::::::
colord:!:19871::::::
nm-openvpn:!:19871::::::
nm-openconnect:!:19871::::::
_galera:!:19871::::::
mysql:!:19871::::::
stunnel4:!*:19871::::::
_rpc:!:19871::::::

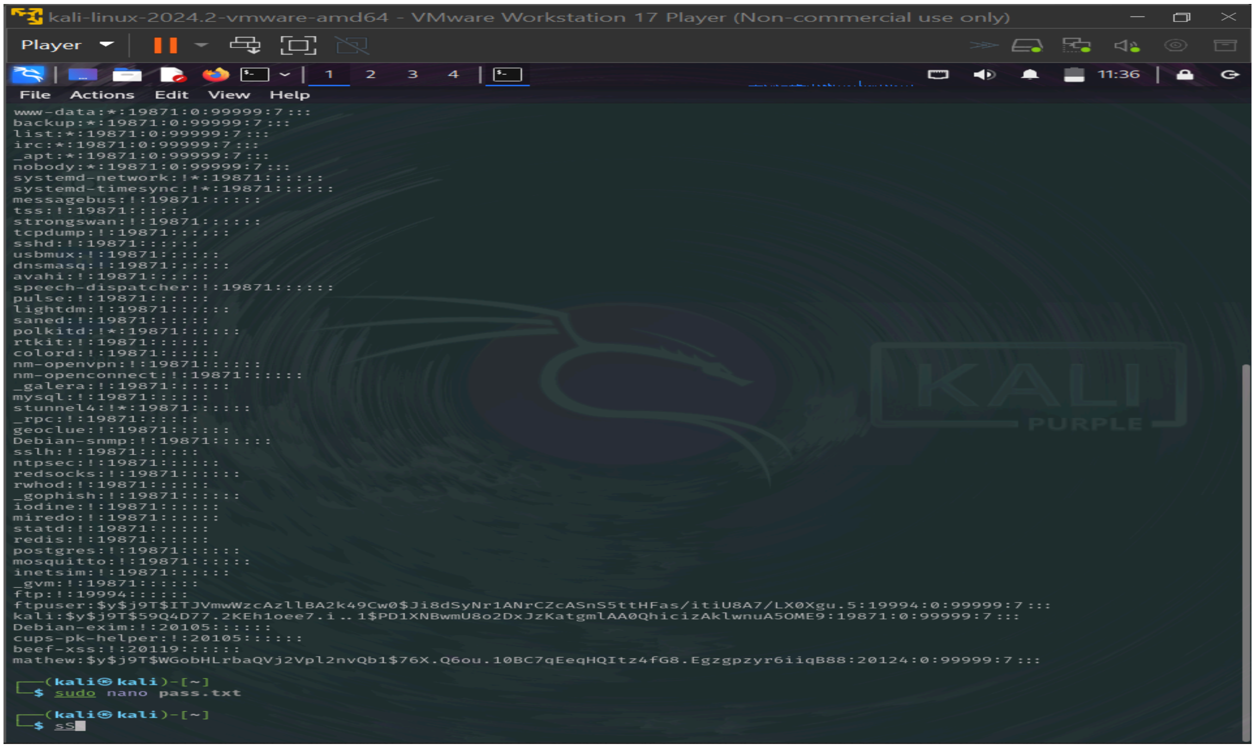

kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
10:17
File Actions Edit View Help
news:*:19871:0:99999:7:::
uucp:*:19871:0:99999:7:::
proxy:*:19871:0:99999:7:::
www-data:*:19871:0:99999:7:::
backup:*:19871:0:99999:7:::
list:*:19871:0:99999:7:::
irc:*:19871:0:99999:7:::
_apt:*:19871:0:99999:7:::
nobody:*:19871:0:99999:7:::
systemd-network:!*:19871::::::
systemd-timesync:!*:19871::::::
messagebus:!:19871::::::
tss:!:19871::::::
strongswan:!:19871::::::
tcpdump:!:19871::::::
sshd:!:19871::::::
usbmux:!:19871::::::
dnsmasq:!:19871::::::
avahi:!:19871::::::
speech-dispatcher:!:19871::::::
pulse:!:19871::::::
lightdm:!:19871::::::
saned:!:19871::::::
polkitd:!*:19871::::::
rtkit:!:19871::::::
colord:!:19871::::::
nm-openvpn:!:19871::::::
nm-openconnect:!:19871::::::
_galera:!:19871::::::
mysql:!:19871::::::
stunnel4:!*:19871::::::
_rpc:!:19871::::::
geoclue:!:19871::::::
Debian-snmp:!:19871::::::
sslh:!:19871::::::
ntpsec:!:19871::::::
redsocks:!:19871::::::
rwhod:!:19871::::::
_gophish:!:19871::::::
iodine:!:19871::::::
miredo:!:19871::::::
statd:!:19871::::::
redis:!:19871::::::
postgres:!:19871::::::
mosquitto:!:19871::::::
inetsim:!:19871::::::
_gvm:!:19871::::::
ftp:!:19994::::::
ftpuser:yj9T$ITJVmwWzcAzllBA2k49Cw0$Ji8dSyNr1ANrCZcASnS5ttHFas/itiU8A7/LX0Xgu.5:19994:0:99999:7:::
kali:yj9T$59Q4D77.2KEh1oee7.i..1$PD1XNBwmU8o2DxJzKatgmlAA0QhicizAklwnuA5OME9:19871:0:99999:7:::
Debian-exim:!:20105::::::
cups-pk-helper:!:20105::::::
beef-xss:!:20119::::::
mathew:yj9T$Y6JGnqXTK.7Zfl4nOtVUY0$j96LhzG9VjmVBQ6y2D22mKD9I1pIpAJIHJONDV.WH6A:20124:0:99999:7:::
┌──(kali㉿kali)-[~]
└─$

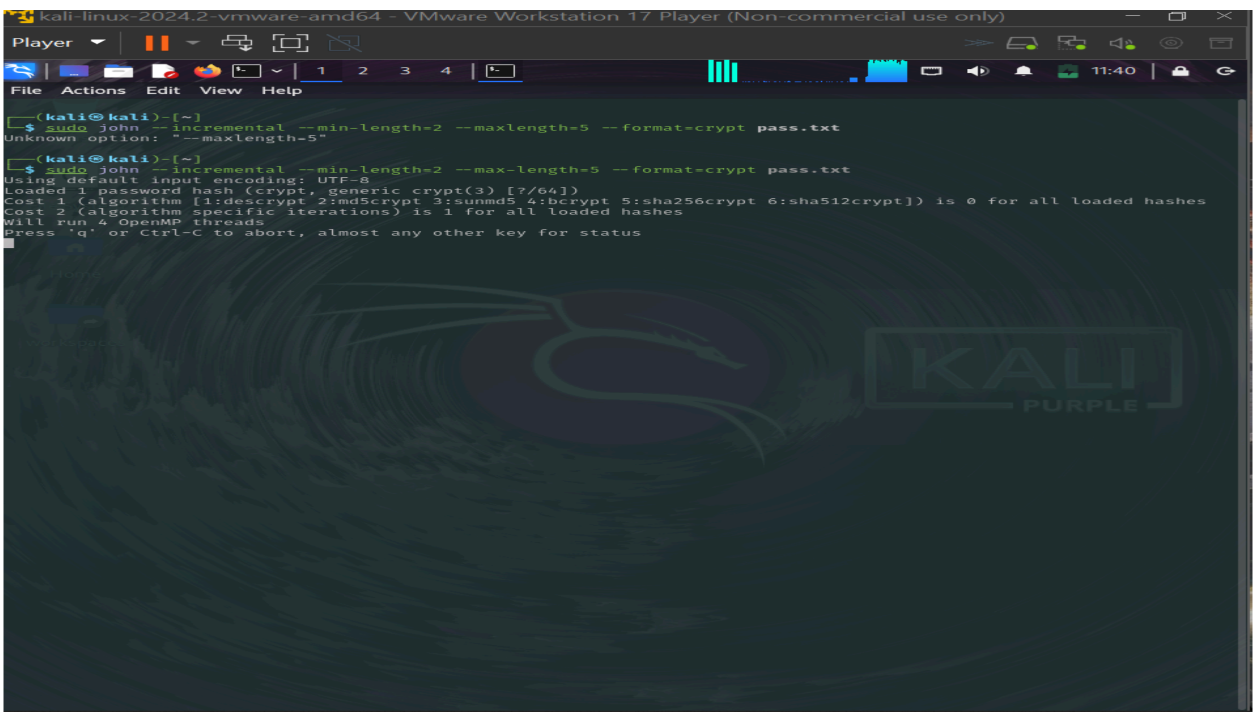

kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
10:33
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo john --single --format=crypt pass.txt
Created directory: /root/.john
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
MaThEw (mathew)
1g 0:00:00:19 DONE (2025-02-05 10:33) 0.05117g/s 63.86p/s 63.86c/s 63.86C/s 99Mathew999..m9999990
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
┌──(kali㉿kali)-[~]
└─$
KALI
PURPLE

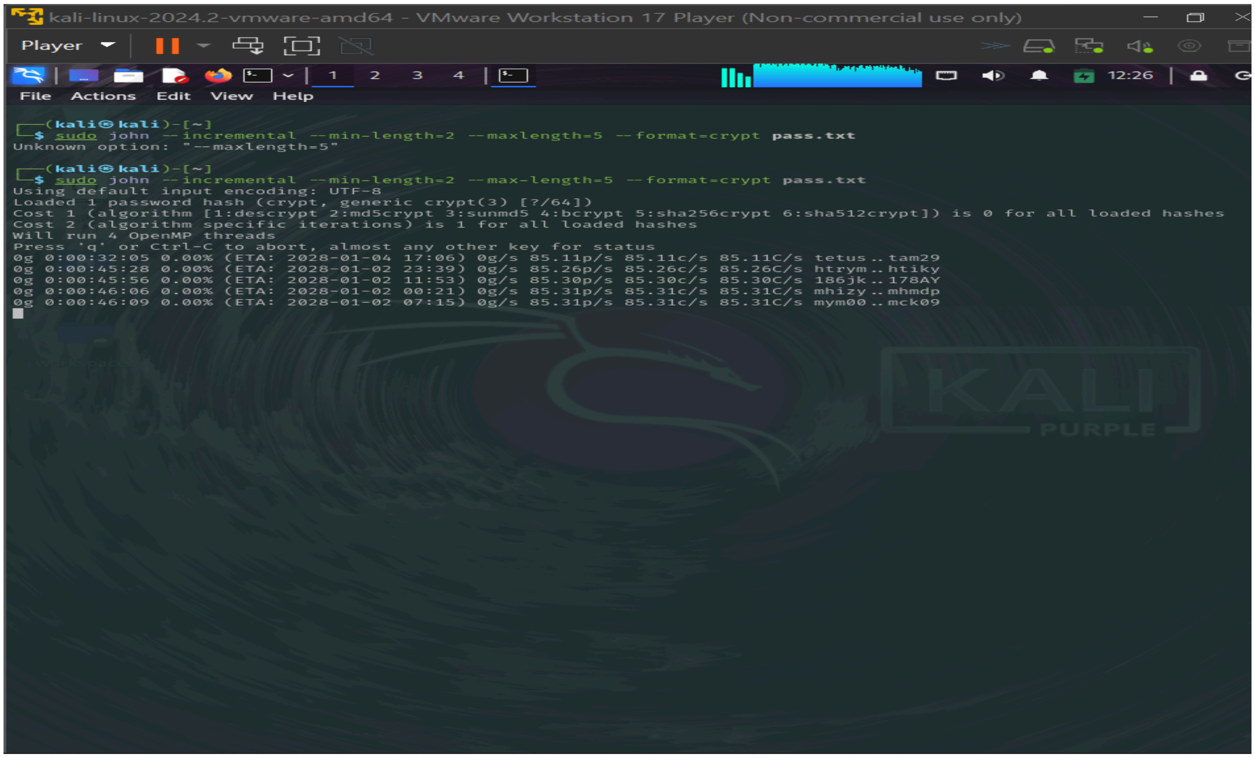

kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
10:37
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ cd /usr/share/wordlists
┌──(kali㉿kali)-[/usr/share/wordlists]
└─$ ls
amass dirbuster fasttrack.txt hydra.restore legion rockyou.txt wfuzz
dirb dnsmap.txt fern-wifi john.lst nmap.lst sqlmap.txt wifite.txt
┌──(kali㉿kali)-[/usr/share/wordlists]
└─$ sudo cat fasttrack.txt
Spring2017
Spring2021
spring2021
Summer2021
summer2021
Autumn2021
autumn2021
Fall2021
fall2021
Winter2021
winter2021
Spring2020
spring2020
Summer2020
summer2020
Autumn2020
autumn2020
Fall2020
fall2020
Winter2020
winter2020
Spring2019
spring2019
Summer2019
summer2019
Autumn2019
autumn2019
Fall2019
fall2019
Winter2019
winter2019
Spring2018
spring2018
Summer2018
summer2018
Autumn2018
autumn2018
Fall2018
fall2018
Winter2018
winter2018
Spring2016
Spring2015
Spring2014
Spring2013
spring2017
KALI
PURPLE

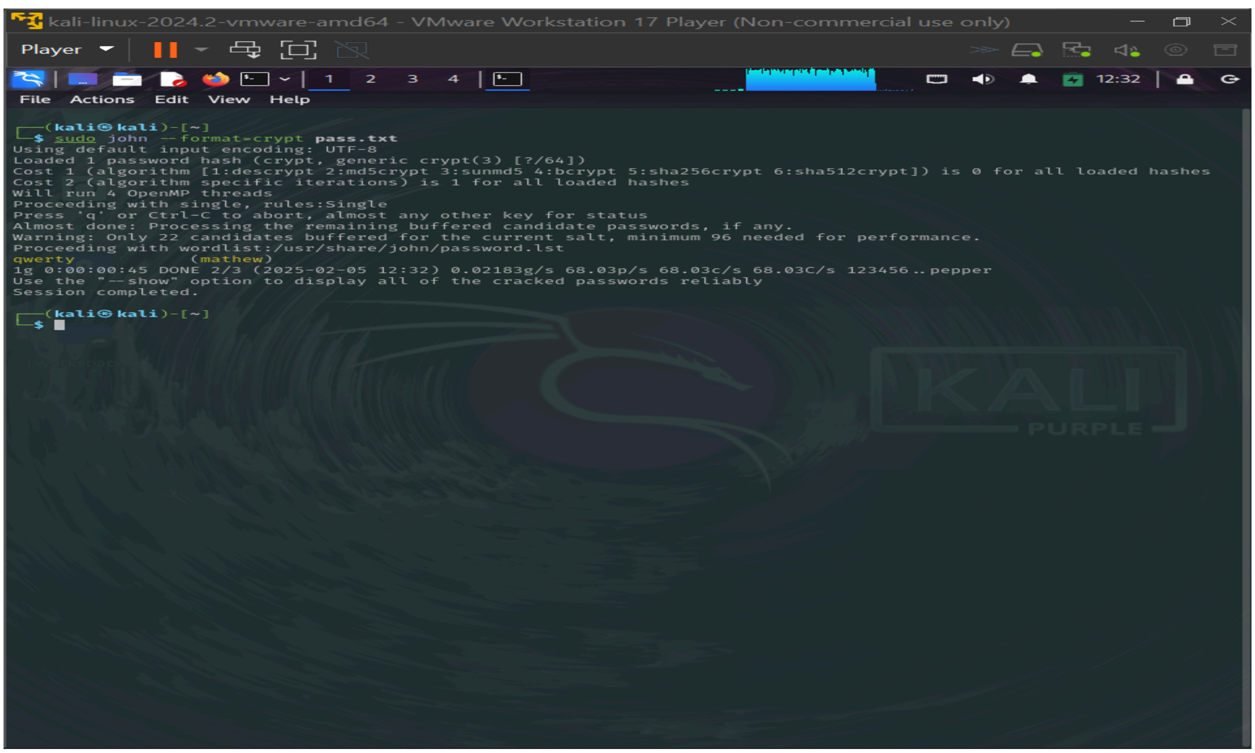

kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
10:49
File Actions Edit View Help
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:992:992:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:100:102::/nonexistent:/usr/sbin/nologin
tss:x:101:104:TPM software stack,,,:/var/lib/tpm:/bin/false
strongswan:x:102:65534::/var/lib/strongswan:/usr/sbin/nologin
tcpdump:x:103:105::/nonexistent:/usr/sbin/nologin
sshd:x:104:65534::/run/sshd:/usr/sbin/nologin
usbmux:x:105:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:106:108:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:107:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
pulse:x:108:110:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
lightdm:x:109:112:Light Display Manager:/var/lib/lightdm:/bin/false
saned:x:110:114::/var/lib/saned:/usr/sbin/nologin
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
rtkit:x:111:115:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:112:116:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
nm-openvpn:x:113:117:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
nm-openconnect:x:114:118:NetworkManager OpenConnect plugin,,,:/var/lib/NetworkManager:/usr/sbin/nologin
_galera:x:115:65534::/nonexistent:/usr/sbin/nologin
mysql:x:116:120:MariaDB Server,,,:/nonexistent:/bin/false
stunnel4:x:990:990:stunnel service system account:/var/run/stunnel4:/usr/sbin/nologin
_rpc:x:117:65534::/run/rpcbind:/usr/sbin/nologin
geoclue:x:118:122::/var/lib/geoclue:/usr/sbin/nologin
Debian-snmp:x:119:123::/var/lib/snmp:/bin/false
sslh:x:120:124::/nonexistent:/usr/sbin/nologin
ntpsec:x:121:127::/nonexistent:/usr/sbin/nologin
redsocks:x:122:128::/var/run/redsocks:/usr/sbin/nologin
rwhod:x:123:65534::/var/spool/rwho:/usr/sbin/nologin
_gophish:x:124:130::/var/lib/gophish:/usr/sbin/nologin
iodine:x:125:65534::/run/iodine:/usr/sbin/nologin
miredo:x:126:65534::/var/run/miredo:/usr/sbin/nologin
statd:x:127:65534::/var/lib/nfs:/usr/sbin/nologin
redis:x:128:131::/var/lib/redis:/usr/sbin/nologin
postgres:x:129:132:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
mosquitto:x:130:133::/var/lib/mosquitto:/usr/sbin/nologin
inetsim:x:131:134::/var/lib/inetsim:/usr/sbin/nologin
_gvm:x:132:135::/var/lib/openvas:/usr/sbin/nologin
ftp:x:133:138:ftp daemon,,,:/srv/ftp:/usr/sbin/nologin
ftpuser:x:1001:1001:california,21,123456789,234567891,45673:/home/ftpuser:/bin/bash
kali:x:1000:1000:CALLEDALIA,,,:/home/kali:/usr/bin/zsh
Debian-exim:x:134:139::/var/spool/exim4:/usr/sbin/nologin
cups-pk-helper:x:135:140:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
beef-xss:x:136:141::/var/lib/beef-xss:/usr/sbin/nologin
mathew:x:1002:1002:mathew,0,0,0,0:/home/mathew:/bin/bash
mather:x:1003:1003:0,0,0,0,0:/home/mather:/bin/bash
(kali㉿kali)-[~]
$


kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
11:06
File Actions Edit View Help
(kali㉿kali)-[~]
$ sudo nano pass.txt
(kali㉿kali)-[~]
$ sudo john --wordlist=/usr/share/wordlists/fasttrack.txt --format=crypt pass.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:03 DONE (2025-02-05 11:04) 0g/s 76.16p/s 76.16c/s 76.16C/s Summer2010..starwars
Session completed.
(kali㉿kali)-[~]
$


kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
11:36
File Actions Edit View Help
www-data:*:19871:0:99999:7:::
backup:*:19871:0:99999:7:::
list:*:19871:0:99999:7:::
irc:*:19871:0:99999:7:::
_apt:*:19871:0:99999:7:::
nobody:*:19871:0:99999:7:::
systemd-network:!*:19871::::::
systemd-timesync:!*:19871::::::
messagebus:!:19871::::::
tss:!:19871::::::
strongswan:!:19871::::::
tcpdump:!:19871::::::
sshd:!:19871::::::
usbmux:!:19871::::::
dnsmasq:!:19871::::::
avahi:!:19871::::::
speech-dispatcher:!:19871::::::
pulse:!:19871::::::
lightdm:!:19871::::::
saned:!:19871::::::
polkitd:!*:19871::::::
rtkit:!:19871::::::
colord:!:19871::::::
nm-openvpn:!:19871::::::
nm-openconnect:!:19871::::::
_galera:!:19871::::::
mysql:!:19871::::::
stunnel4:!*:19871::::::
_rpc:!:19871::::::
geoclue:!:19871::::::
Debian-snmp:!:19871::::::
sslh:!:19871::::::
ntpsec:!:19871::::::
redsocks:!:19871::::::
rwhod:!:19871::::::
_gophish:!:19871::::::
iodine:!:19871::::::
miredo:!:19871::::::
statd:!:19871::::::
redis:!:19871::::::
postgres:!:19871::::::
mosquitto:!:19871::::::
inetsim:!:19871::::::
_gvm:!:19871::::::
ftp:!:19994::::::
ftpuser:yj9T$ITJVmwWzcAzllBA2k49Cw0$Ji8dSyNr1ANrCZcASnS5ttHFas/itiU8A7/LX0Xgu.5:19994:0:99999:7:::
kali:yj9T$59Q4D77.2KEh1oee7.i..1$PD1XNBwmU8o2DxJzKatgmlAA0QhicizAklwnuA5OME9:19871:0:99999:7:::
Debian-exim:!:20105::::::
cups-pk-helper:!:20105::::::
beef-xss:!:20119::::::
mathew:yj9T$WGobHLrbaQVj2Vpl2nvQb1$76X.Q6ou.10BC7qEeqHQItz4fG8.Egzgpzyr6iiqB88:20124:0:99999:7:::
┌──(kali㉿kali)-[~]
└─$ sudo nano pass.txt
┌──(kali㉿kali)-[~]
└─$ sS


kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
1 2 3 4
11:40
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo john --incremental --min-length=2 --maxlength=5 --format=crypt pass.txt
Unknown option: "--maxlength=5"
┌──(kali㉿kali)-[~]
└─$ sudo john --incremental --min-length=2 --max-length=5 --format=crypt pass.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status


kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
12:26
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo john --incremental --min-length=2 --maxlength=5 --format=crypt pass.txt
Unknown option: "--maxlength=5"
┌──(kali㉿kali)-[~]
└─$ sudo john --incremental --min-length=2 --max-length=5 --format=crypt pass.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:32:05 0.00% (ETA: 2028-01-04 17:06) 0g/s 85.11p/s 85.11c/s 85.11C/s tetus..tam29
0g 0:00:45:28 0.00% (ETA: 2028-01-02 23:39) 0g/s 85.26p/s 85.26c/s 85.26C/s htrym..htiky
0g 0:00:45:56 0.00% (ETA: 2028-01-02 11:53) 0g/s 85.30p/s 85.30c/s 85.30C/s 186jk..178AY
0g 0:00:46:06 0.00% (ETA: 2028-01-02 00:21) 0g/s 85.31p/s 85.31c/s 85.31C/s mhizy..mhmdp
0g 0:00:46:09 0.00% (ETA: 2028-01-02 07:15) 0g/s 85.31p/s 85.31c/s 85.31C/s mym00..mck09


kali-linux-2024.2-vmware-amd64 - VMware Workstation 17 Player (Non-commercial use only)
Player
12:32
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo john --format=crypt pass.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 4 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Warning: Only 22 candidates buffered for the current salt, minimum 96 needed for performance.
Proceeding with wordlist:/usr/share/john/password.lst
qwerty (mathew)
1g 0:00:00:45 DONE 2/3 (2025-02-05 12:32) 0.02183g/s 68.03p/s 68.03c/s 68.03C/s 123456..pepper
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
┌──(kali㉿kali)-[~]
└─$

Key Concepts

Password Storage in Linux

When a user is added to a Linux system, their credentials are stored in two critical files:

1. **/etc/passwd**: Contains user information such as username, user ID, group ID, home directory, and default shell.
2. **/etc/shadow**: Stores the username along with the hashed password. The hashing algorithm used here is **bcrypt**, which incorporates a **salt** to enhance security.

Bcrypt Hashing

Bcrypt is a robust hashing strategy that uses multiple rounds of hashing to secure passwords. The number of rounds is denoted by a specific character in the hash, and the length of the hash varies based on the number of rounds. This makes bcrypt resistant to brute-force attacks.

Modes of Operation in John the Ripper

1. **Single Crack Mode**

In this mode, John the Ripper uses the **username** and its variations to guess the password. It generates possible passwords by altering the case of letters in the username and hashing them to compare against the stored hash.

Command:

john --single --format=crypt pass.txt

2. **Wordlist Mode**

John the Ripper uses a predefined list of common passwords (wordlist) to crack the password. It hashes each word in the list and compares it to the target hash. **For the wordlist mode to work, the password must be present in the wordlist.** If you are unsure, it is recommended to use **Incremental Mode** or **Default Mode**.

Command:
john --wordlist=/usr/share/wordlists/fasttrack.txt --format=crypt pass.txt

3. **Incremental Mode**
This mode is the most time-consuming but also the most thorough. It attempts all possible combinations of characters within a specified length range. This mode is useful when other methods fail. **Note that Incremental Mode can take a significant amount of time**, especially for longer passwords.

Command:
john --incremental --min-length=2 --max-length=3 --format=crypt pass.txt
```

4. **Default Mode**
When no specific mode is specified, John the Ripper starts with **Single Crack Mode**, then proceeds to **Wordlist Mode**, and finally resorts to **Incremental Mode** if the previous attempts fail. This mode is a balanced approach that combines the efficiency of Single Crack and Wordlist modes with the thoroughness of Incremental Mode.

**Command:**
john --format=crypt pass.txt

**Hashing Strategies**
John the Ripper supports a variety of hashing strategies. You can search online to find the complete set of hashing formats supported by John the Ripper. The `--format` flag is used to specify the hashing algorithm during the cracking process.

**Demonstration**
```

Adding a User
To demonstrate password cracking, we first add a user with a specific password:

adduser Matthew
```
**Password: `Mathew`**

**Extracting Hash Information**
The user's hashed password is stored in the `/etc/shadow` file. We copy this hash to a separate file (`pass.txt`) for cracking.

**cat /etc/shadow | grep Matthew > pass.txt**
```

Cracking the Password
We then use John the Ripper to crack the password in different modes:

1. **Single Crack Mode:**
john --single --format=crypt pass.txt
```

**2. **Wordlist Mode:****
**john --wordlist=/usr/share/wordlists/fasttrack.txt --format=crypt pass.txt**
```

3. **Incremental Mode:**
john --incremental --min-length=2 --max-length=3 --format=crypt pass.txt
```

**4. **Default Mode:****
**john --format=crypt pass.txt**
```

Conclusion
John the Ripper is an essential tool for cybersecurity professionals, enabling them to test the strength of passwords and identify vulnerabilities in password storage. By understanding and utilizing its different modes, you can efficiently crack passwords and enhance system security.

****Key Takeaways:****
- **Single Crack Mode is useful when the password is derived from the username.**
- **Wordlist Mode is effective if the password is common and present in the wordlist.**
- **Incremental Mode is the most thorough but time-consuming, ideal for complex passwords.**
- **Default Mode combines all strategies for a balanced approach.**

Disclaimer: Password cracking should only be performed on systems where you have explicit permission. Unauthorized access to systems is illegal and unethical.